Spike-Driven Dreamer: Persistent Dendrites Give Spiking World Models Long-Term Memory

Anonymous CVPR submission

Paper ID *****

Abstract

Spiking neural networks (SNNs) compute with sparse binary spikes and run on neuromorphic hardware at a fraction of the energy of dense networks, which makes them a natural choice for embodied agents. Such agents see only part of their world, and a world model gives them memory by compressing past observations into a recurrent state. We find that existing spiking world models forget within a few steps, even when their neurons have long time constants. The cause is reset: membrane potentials are cleared at every environment step and dendrites at every somatic spike, so all memory has to pass through binary spikes at every step. Causal interventions on a spiking world model confirm this reset amnesia and show that long time constants also need a normalized write gain. Building on these findings, we propose the Spike-Driven Dreamer (SD-Dreamer), a spiking world model built on persistent dendritic neurons, in which dendrites store memory and spikes communicate it. Persistent Dendritic Memory (PDM) carries integrator and resonant dendritic channels across environment steps and lets spikes write into them without resetting them, and a Normalized Write Gain (NWG) keeps channels of every retention time on the same scale. The integrators hold what the agent has seen, and the resonators also record when. A stateless soma reads the channels and sends a spike count, the only signal that passes between neurons. On BSuite memory length, SD-Dreamer recalls a cue 101 steps back within the budget of R2I, while DreamerV3 fails beyond 51 steps. On POPGym RepeatPrevious, it is the only model, spiking or not, that learns to name the card shown exactly 32 steps earlier within 2M environment steps, and its world model also recalls exact lags of 64 steps on a task held out from development. Spiking world models thus remember over long horizons once their memory lives in dendrites that spikes do not reset.

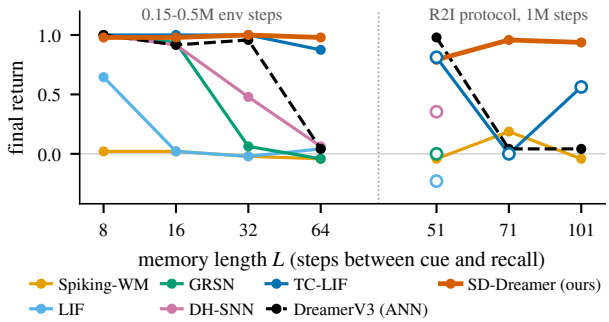


Figure 1. **Spiking world models forget.** Final return on BSuite memory length [32]: the agent sees a cue, then L uninformative steps, and is rewarded for recalling the cue (1 = always correct, 0 = chance). All models share one spiking world model and differ in the recurrent core (Sec. 5.1). Left: budgets of 150k to 500k environment steps. Right: L = 51, 71 and 101 with the lengths and budget of R2I [38] (1M steps). Spiking-WM fails at every length, recurrent LIF neurons stop at 8 steps, and GRSN and DH-SNN at 16; TC-LIF, whose state is carried across steps and never cleared, learns 51 and 101 steps (71: in progress). SD-Dreamer reaches a return of 0.9 at every length up to 101 steps, and DreamerV3 up to 51 (Tab. 1). Open markers: runs in progress.

1. Introduction

Neuromorphic computing brings the structure and dynamics of biological nervous systems into the design of computational models and hardware [37]. Its representative model, the spiking neural network (SNN) [24], processes information through neuron dynamics and communicates by discrete spikes. Sparse activation and event-driven computation make SNNs a basis for energy-efficient inference on neuromorphic hardware [5, 30]. SNNs have recently been applied to image recognition [8, 47, 53], temporal modeling [3, 48, 50, 52] and control through reinforcement learning (RL) [21, 35, 42].

Unlike standard deep learning tasks, RL requires an agent to learn a decision policy from the experience of interacting with its environment. Model-based RL learns the dynamics

of the environment with a world model, predicts the consequences of actions, and uses these predictions to improve the policy [11, 12]. Methods represented by Dreamer [13, 15] integrate past observations and actions into a recurrently updated latent state, generate imagined trajectories in the latent space and learn the policy from them. In partially observable tasks, this state also serves as memory and must retain information relevant to later decisions across many environment steps [26, 32, 33]. World models built on artificial neural networks (ANNs) have strengthened this long-horizon memory with state space models [6, 38]. In SNNs, membrane potentials and dendritic states evolve over time and can retain past information [3, 50, 52]. Building on this, a recent spiking world model uses multicompartment neurons to reduce inference energy substantially while approaching the performance of ANN world models [41].

However, existing spiking world models still face two challenges on long-horizon memory tasks: writing early observations effectively into neuron states with long time constants, and preserving this information through continued interaction and spiking until later decisions need it. In the standard Euler update, the write gain of $1/\tau$, tied to the time constant, makes compartments with long time constants respond too weakly to input to drive spikes. Meanwhile, several spiking RL methods and world models clear the membrane potentials at the start of every environment step [21, 41, 42], which confines the memory held by the membrane potential to a single step. In multicompartment neurons, a somatic spike also resets the dendritic state [41], so a dendritic memory trace survives only until the neuron fires again. We term the memory failure caused by these mechanisms reset amnesia. As a result, even with longer time constants, existing spiking world models are limited in forming and retaining long-term memory and cannot carry key information from early observations to later decisions (Fig. 1).

To address these problems, we propose the Spike-Driven Dreamer (SD-Dreamer), a spiking world model whose recurrent core is built from a newly designed Persistent Dendritic Neuron (PDN). As shown in Fig. 2, a PDN separates memory storage from communication between neurons. First, we propose Persistent Dendritic Memory (PDM): integrator channels store the content of past observations, resonant channels [19] encode the time elapsed since an observation, and both keep their state across environment steps and through somatic spikes, so this information persists. Next, we design a Normalized Write Gain (NWG), which scales the write of spike input by the retention time of each channel; it compensates the input attenuation of channels with long retention times, so the memory they store can drive the soma. Finally, a stateless soma reads this dendritic memory and emits a spike count, passing information between neurons while the memory stays inside the neuron. We evaluate the long-horizon memory of SD-Dreamer on several RL tasks, with vector observations and first-person images of 3D mazes. The results show that SD-Dreamer outperforms the compared spiking baselines overall on long-horizon memory tasks, including models with recent spiking neurons and an existing spiking world model. Specifically, within the training budget of R2I [38], SD-Dreamer acts on a cue seen 101 steps earlier in BSuite, whereas DreamerV3 fails beyond 51 steps. On POPGym RepeatPrevious Medium, which requires recalling the card shown exactly 32 steps earlier, SD-Dreamer is the only compared model whose return rises above chance within 2M environment steps. World model probes and ablations further confirm the roles of PDM and NWG in long-horizon memory and show that exact recall of the observation at a specific past step depends on the resonant channels. We further evaluate SD-Dreamer on the standard control tasks of DMC [43], where its returns from both proprioception and pixels are comparable to those of the spiking baselines. These results show that SD-Dreamer offers an effective way to build spiking world models with long-horizon memory. The main contributions are as follows:

- We identify and analyze reset amnesia in spiking world models, show how the step reset, the spike reset and a write gain tied to the time constant limit the formation and retention of long-term memory, and confirm their roles with causal interventions.
- We propose Persistent Dendritic Memory (PDM) and a Normalized Write Gain (NWG). PDM carries the state of integrator and resonant channels across environment steps and keeps it through somatic spikes, storing both the content and the timing of past observations; NWG scales the write by the retention time to compensate the input attenuation of channels with long retention times. Together with a stateless soma, they form the Persistent Dendritic Neuron (PDN), which separates memory storage from spike communication.
- Building on PDN, we develop SD-Dreamer, evaluate its long-horizon memory on BSuite memory length, POPGym RepeatPrevious and Memory Maze, and verify the roles of PDM, NWG and the resonant channels with world model probes and ablations. Its long-horizon memory surpasses the compared spiking baselines overall, and on DMC its returns from proprioception and pixels are comparable to those of the spiking baselines.

2. Related Work

World models and memory. Latent world models learn a recurrent state from observations and actions and train behavior in imagination [11–15]. In partially observable tasks this state is the agent's memory, and the recurrent core sets how long it lasts: a GRU in the Dreamer family, attention over a finite context window in transformer world

models [25, 36, 51], and a linear state space recurrence in S4WM [6] and R2I [38]. With this recurrence, R2I solves memory tasks that DreamerV3 cannot on BSuite memory length [32], POPGym [26] and Memory Maze [33], and linear recurrences with decaying and rotating state also serve as memory cells in model-free RL [23, 27, 31, 44]. Probes of the learned state measure this memory apart from exploration and credit assignment [29, 33, 45]. All these models are dense ANNs whose units pass graded state vectors from step to step. In a spiking world model, neurons pass only binary spikes, so long-horizon memory has to be held inside the neurons themselves.

Spiking reinforcement learning and world models. Spiking actors and Q-networks [21, 42] run a feedforward SNN for a few simulation steps per environment step and clear it before the next step, which confines their state to a single environment step. Recurrent spiking agents carry information across steps: GRSN [35] aligns a gated recurrent spiking neuron with environment time for model-free RL in POMDPs, and recurrent LIF networks have been trained end to end for model-based control [17]. Spiking-WM [41] brings multicompartment neurons into the world model of DreamerV3 and evaluates it on DMC from pixels and on speech classification. Its neurons are cleared at every environment step and its dendrites at every somatic spike, so their time constants act only within one step, and changing them barely changes its performance [41]. These works evaluate control and classification; inside a world model on memory benchmarks, their cores remember at most 16 steps (Sec. 5.2), and Sec. 3.3 traces this failure to reset amnesia.

Spiking neurons with long memory. Neuron models extend the memory of a spiking neuron with adaptive thresholds [3, 48], two compartments [50], dendritic branches with heterogeneous time constants [52] and resonate-and-fire dynamics [16, 19, 49]. Removing the reset makes the membrane linear and parallelizable [10], and spiking state space layers attach spiking outputs to linear recurrences [2, 7, 39, 40]. These models are trained and tested on offline sequence classification, where the network sees each sequence in full and is reset between samples. A world model instead filters an unbounded stream one environment step at a time and continues its state in imagination; we run DH-SNN and TC-LIF in this setting as baselines. SD-Dreamer is designed for this setting: its memory persists across environment steps in dendrites that no spike resets, its retention times are counted in environment steps, and its soma holds no state. PDNs borrow two properties of biological neurons: dendrites act as computational subunits [4, 20, 22, 34], and many neurons resonate below threshold [18].

3. Preliminaries

3.1. Recurrent world models

Our world model is the recurrent state space model (RSSM) [12] of DreamerV3 [15]. An RSSM learns the dynamics of an environment in a compact latent space, and the agent learns its policy from trajectories imagined in this space; with one fixed configuration, DreamerV3 learns more than 150 tasks, from continuous control to Atari and Minecraft [15]. At environment step t, the sequence model updates the deterministic state h_t, the encoder infers the stochastic latent z_t from h_t and the observation o_t, and the dynamics predictor predicts z_t from h_t alone:

$$\begin{aligned} h_t &= f_\phi(h_{t-1}, z_{t-1}, a_{t-1}), \\ z_t &\sim q_\phi(z_t \mid h_t, o_t), \quad \hat{z}_t \sim p_\phi(\hat{z}_t \mid h_t). \end{aligned} \tag{1}$$

Reward, continuation and decoder heads read (h_t, z_t). The sequence model f_ϕ, a GRU in DreamerV3, is the only path from one step to the next, so in a partially observable task h_t is the memory of the agent, both when the model filters observations and when it imagines.

3.2. Spiking neurons and spiking world models

SNNs compute with neurons that integrate synaptic input into a membrane potential and communicate by binary spikes [24]. The leaky integrate-and-fire (LIF) neuron, discretized with the Euler method [9], charges, fires and resets in simulation step n as

$$\begin{aligned} \tilde{u}[n] &= u[n-1] + \frac{1}{\tau}\big(I[n] - u[n-1]\big), \\ s[n] &= \mathrm{H}\big(\tilde{u}[n] - \vartheta\big), \quad u[n] = \tilde{u}[n]\big(1 - s[n]\big), \end{aligned} \tag{2}$$

where $I[n]$ is the synaptic current of the input spikes, τ is the membrane time constant, $\tilde{u}$ and u are the potential before and after the reset, ϑ is the threshold and H is the Heaviside step, trained with a surrogate derivative [28, 46].

Spiking-WM [41] builds the RSSM of Eq. (1) from LIF layers that run $T = 5$ simulation steps per environment step, and replaces the GRU with multicompartment neurons (MCNs), so that $h_t \in \{0,1\}^{T\times N}$ is the spike output of the sequence model. An MCN has a basal dendrite, an apical dendrite and a soma. Each dendrite $c \in \{\mathrm{b}, \mathrm{a}\}$ integrates its synaptic current with its own time constant τ_c, and the soma integrates the basal potential through a gate set by the apical potential:

$$v^c[n] = v^c[n-1] + \frac{1}{\tau_c}\big(I^c[n] - v^c[n-1]\big), \tag{3}$$

$$\tilde{u}[n] = u[n-1] + \frac{\sigma(v^{\mathrm{a}}[n])}{\tau}\big(v^{\mathrm{b}}[n] + I^{\mathrm{s}}[n] - u[n-1]\big),$$

where $I^c[n] = W^c x_t[n] + W^{h,c} h_{t-1}[n]$ combines the input spikes $x_t[n]$, which encode (z_{t-1}, a_{t-1}), with the spikes

of the sequence model at the previous environment step, $I^s[n] = W^s x_t[n]$, and σ is the sigmoid. The soma fires and resets as in Eq. (2). Two resets act on this neuron: the step reset sets all compartments to zero at the start of every environment step, and the spike reset clears the soma and both dendrites whenever the soma fires.[1]

3.3. Reset amnesia

A spiking core passes information from one environment step to the next through the spikes h_{t-1} it feeds back and through the states of its neurons, and only the neuron states can hold a graded memory over many steps. We call the loss of this memory caused by the resets and the write gain of existing spiking neurons reset amnesia.

The step reset, used by spiking agents [21, 41, 42], removes the neuron states between environment steps. The core then reduces to a binary recurrent network $h_t = \Phi(x_t, h_{t-1})$ that must encode everything it knows about the past again into the $T \times N$ bits of h_{t-1} at every step, and the gradient from a reward L steps after a cue crosses L surrogate derivatives. The time constants act only within the T simulation steps of one environment step, which is why changing them barely changes the performance of Spiking-WM [41].

The spike reset of the LIF neuron and of the MCN truncates a trace at the next spike. For a compartment with this reset, the influence of a past input on the potential is

$$\frac{\partial \tilde{u}[n]}{\partial I[n-j]} = \frac{1}{\tau}\Big(1-\frac{1}{\tau}\Big)^j \prod_{i=1}^{j} \big(1 - s[n-i]\big), \qquad (4)$$

which vanishes once the neuron has fired, so the memory of the compartment lasts at most until its next spike. Even without resets, the Euler updates of Eqs. (2) and (3) write the input with the gain $1/\tau$: for an input of unit variance that is uncorrelated over time, the potential has the stationary variance $1/(2\tau - 1) \approx 1/(2\tau)$, so a compartment with a long time constant keeps a cue but stores it too faintly to drive spikes.

Reset amnesia thus gives a design rule: the memory of a spiking world model must reside in a state that persists across environment steps, is not reset by spikes, and is written with a gain that does not vanish for long time constants. Section 5.5 tests each mechanism with interventions on Spiking-WM.

4. Spike-Driven Dreamer

SD-Dreamer follows the design rule of Sec. 3.3: it keeps the spiking world model of Sec. 3.2 and replaces its recurrent core with two layers of persistent dendritic neurons (PDNs), as shown in Fig. 2. The second layer reads the spikes of the first, and its spikes form the deterministic state h_t; the encoder, the latent, the heads and the actor-critic remain LIF networks trained with the losses of DreamerV3 [15].

A PDN separates memory from communication: dendritic channels store the memory, and a stateless soma communicates it by spikes. A layer has N neurons with M dendritic channels each. At environment step t it receives the firing rate x_t of its input spikes over the T simulation steps and its own firing rate $\bar{s}_{t-1}$ of the previous step, and computes

$$\begin{aligned} I_t &= \mathrm{LN}\big(B x_t + R\,\bar{s}_{t-1}\big), \\ (d_t, \tilde{d}_t) &= \mathrm{PDM}\big(d_{t-1}, \tilde{d}_{t-1}; I_t\big), \\ s_t &= \mathrm{Soma}\big(d_t, \tilde{d}_t, x_t\big), \end{aligned} \qquad (5)$$

where B and R are synaptic matrices, LN is layer normalization with learned scale and bias [1], $I_t \in \mathbb{R}^{NM}$ is the synaptic current into the dendrites, $(d_t, \tilde{d}_t)$ is the state of the dendritic channels and s_t are the T binary spike frames of the layer. Persistent Dendritic Memory (Sec. 4.1) carries the dendritic state across environment steps and somatic spikes, the Normalized Write Gain (Sec. 4.2) writes the current into channels of every retention time on one scale, and the stateless soma (Sec. 4.3) turns the state into spike counts. All operations act elementwise on the neurons and channels of a layer.

4.1. Persistent Dendritic Memory

In Spiking-WM, the step reset clears all compartments at every environment step and the spike reset clears the dendrites at every somatic spike, so no trace outlasts the next step (Sec. 3.3). Persistent Dendritic Memory (PDM) keeps the dendritic state through both. Each channel has a retention time τ, counted in environment steps, a decay $\lambda = e^{-1/\tau}$ and a phase θ, and its state is a damped rotation driven by the current:

$$\begin{pmatrix} d_t \\ \tilde{d}_t \end{pmatrix} = \lambda\, Q(\theta) \begin{pmatrix} d_{t-1} \\ \tilde{d}_{t-1} \end{pmatrix} + g(\lambda) \begin{pmatrix} I_t \\ 0 \end{pmatrix}, \qquad (6)$$

where $Q(\theta)$ is the rotation by the angle θ and $g(\lambda)$ is the write gain of Sec. 4.2. An integrator channel has $\theta = 0$ and $\tilde{d} \equiv 0$, so $d_t = \lambda d_{t-1} + g(\lambda) I_t$ is a leaky trace; a resonant channel is a damped oscillator, as in resonate-and-fire neurons [19].

Equation (6) is the only update of the channels: they keep their state across environment steps and through somatic spikes, the spikes of the layer enter only as current through R, and the state is cleared only at the start of an episode. The influence of a current k steps back, $g(\lambda)\lambda^k Q(k\theta)$, therefore does not depend on what the neuron fired in between, and the gradient along this path crosses no surrogate derivative. The two kinds of channels store different aspects of the past. An integrator stores what was seen, with inputs of all ages

[1] The equations of [41] show only the somatic reset; the dendritic reset is part of its official implementation (`MCNode.calc_spike`), which all our Spiking-WM runs use.

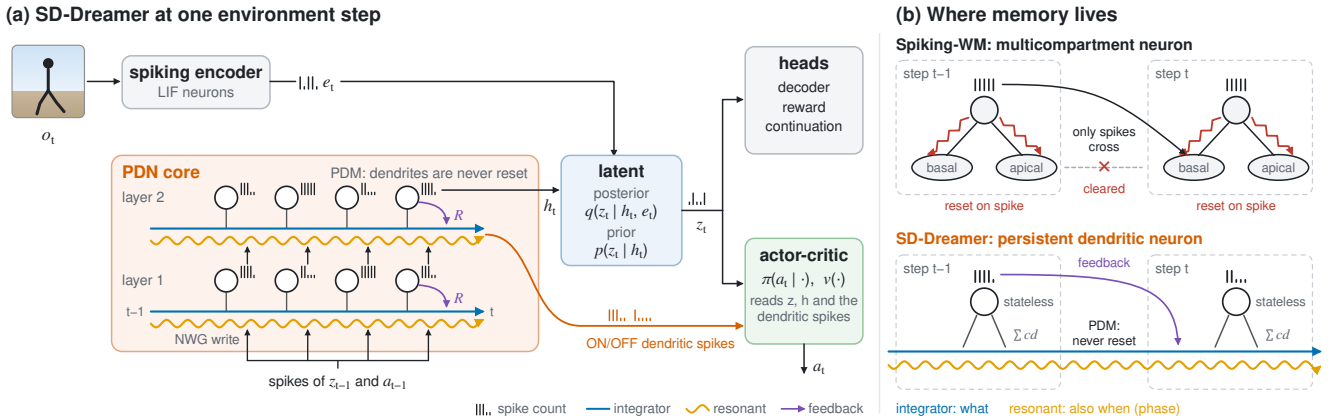


Figure 2. **Spike-Driven Dreamer.** (a) Two layers of persistent dendritic neurons (PDNs) form the recurrent core of a spiking Dreamer. Each neuron has an integrator (blue) and a resonant (orange) dendritic channel; PDM carries them across environment steps without reset, and NWG writes into them on one scale for every retention time. Its stateless soma sends a spike count over the $T = 5$ simulation steps, and the spikes of a layer also feed back into its dendrites (R). The actor and critic also read the dendrites of the second layer as ON and OFF spikes. (b) In Spiking-WM, all compartments are cleared at every step and the dendrites at every somatic spike, so only spikes reach the next step. In a PDN, PDM keeps the dendritic channels across steps; spikes add to them but never clear them.

superposed under the smoothly decaying weights λ^k. A resonator also stores when: an input of age k appears at phase $k\theta$, which separates it from inputs of age $k \pm 1$.

4.2. Normalized Write Gain

Persistence alone does not make long retention times useful, since the gain with which a channel stores its input must not vanish as τ grows. The Euler gain $g = 1 - \lambda \approx 1/\tau$ gives the state a variance of about $1/(2\tau)$ (Sec. 3.3), so channels with long retention times store their input too faintly to drive the soma. The Normalized Write Gain (NWG) sets

$$g(\lambda) = \sqrt{1 - \lambda^2}, \tag{7}$$

the normalization of the LRU [31]. For a normalized current of unit variance, the state then has a variance of $(1 - \lambda^2) \sum_{k \geq 0} \lambda^{2k} = 1$ for every τ. With all channels on one scale, one layer can hold a wide spectrum of retention times: we initialize them uniformly on a log scale in $[1, 1000]$ environment steps and learn them, so no time constant is tuned for a task.

4.3. Stateless soma and policy input

The soma reads the channels of its neuron and the direct input,

$$u_t = \sum_{m=1}^{M} \left(c_m \odot d_{t,m} + \tilde{c}_m \odot \tilde{d}_{t,m} \right) + Dx_t, \tag{8}$$

where $c_m, \tilde{c}_m \in \mathbb{R}^N$ are weights of each neuron and D is a synaptic matrix, and in simulation step $n = 1, \ldots, T$ it emits $s_t^n = \mathrm{H}\big(\mathrm{LN}(u_t) - n\vartheta/T\big)$ with $\vartheta = 0.5$. The soma keeps no state between environment steps, so no reset touches the memory, and its spike count encodes u_t in $T + 1$ levels instead of the single bit of one spike per step. These binary frames are the only signal the neuron sends; the next layer and the feedback read their rate $\bar{s}_t$. The synaptic operations Bx_t, $R\bar{s}_{t-1}$ and Dx_t thus act on spikes, and the dendritic update is local to each neuron, with $O(NM)$ operations per step against $O(N^2)$ for the dense recurrent matrix of a GRU.

Policy input. The actor and critic read z_t, h_t and the dendritic state of the second layer, and the Memory Maze probes read the same features. The dendritic state leaves the neurons as spikes: each state variable, standardized over the layer, is compared with T positive and T negative thresholds $\pm(n - \frac{1}{2})\frac{2}{T}$, $n = 1, \ldots, T$, which gives $2T$ ON and OFF spike frames per environment step. Supp. E compares a policy that reads (z_t, h_t) only.

Training. The dendritic state is carried through training sequences and imagination rollouts and cleared only at episode starts, and all Heaviside steps are trained with the surrogate derivative of the world model.

5. Experiments

5.1. Setup

Models. All models share one spiking world model, the network of Spiking-WM [41] built on DreamerV3 at the size of DreamerV3-S (512 units, $T = 5$), and differ in the recurrent core: **Spiking-WM** (multicompartment neurons with both resets, $\tau = 2$); **recurrent LIF** neurons, cleared at every environment step as usual in spiking RL; **GRSN** [35], gated recurrent spiking neurons; **DH-SNN** [52], LIF neurons with eight dendritic branches of learned time constants;

Table 1. **Memory in reinforcement learning.** BSuite memory length [32]: success as in R2I [38], i.e. whether the evaluation return reaches 0.9 within the budget (✓) or not (✗); steps and returns in Supp. A. †Lengths, budget and training windows of R2I (1M steps; reward head with one output per action). POPGym RepeatPrevious [26] with the environment settings of R2I's code: max-mean episodic return, the metric of POPGym (chance $= -0.5$). n/a: not run, as the model fails at a shorter length. $^{d}agger$R2I with its official code and configuration for 2M steps, measured on its training episodes (for our models the two measures differ by at most 0.07). Blue: in progress.

	BSuite memory length: success							RepeatPrevious: return ↑	
	L=8	16	32	64	51†	71†	101†	Easy	Medium
Spiking-WM [41]	✗	✗	✗	✗	✗	✗	✗	0.64	-0.46
Recurrent LIF	✓	✗	✗	✗	·	n/a	n/a	0.74	-0.44
GRSN [35]	✓	✓	✗	✗	·	n/a	n/a	0.37	-0.47
DH-SNN [52]	✓	✓	✗	✗	·	n/a	n/a	-0.44	-0.46
TC-LIF [50]	✓	✓	✓	✓	✓	·	✓	-0.28	-0.48
DreamerV3 (ANN) [15]	✓	✓	✓	✗	✓	✗	✗	0.42	-0.41
R2I (ANN) [38]	reported in [38], see text							-0.48‡	-0.46‡
SD-Dreamer (ours)	✓	✓	✓	✓	✓	✓	✓	0.72	0.48
budget (env. steps)	150k	200k	300k	500k	1M	1M	1M	2M	2M

TC-LIF [50], two-compartment LIF neurons for long sequences (GRSN, DH-SNN and TC-LIF update once per environment step and carry their state); **DreamerV3** [15], the same architecture without spikes (GRU core, SiLU, one simulation step); and **SD-Dreamer** with two PDN layers of $N = 512$ neurons with $M = 2$ channels, one integrator and one resonant. R2I [38], a DreamerV3 agent with a state space recurrence and the state of the art among ANN world models on memory tasks, runs with its official code and configuration on POPGym; on BSuite we compare with its published results. Within a benchmark, all models use identical hyperparameters, budgets and evaluation. The actor and critic of every model read (z_t, h_t); those of SD-Dreamer also read its dendritic state as spikes (Sec. 4.3), with 4.1M instead of 3.4M parameters (sizes in Supp. A, an actor-critic of 3.4M in Supp. E).

Benchmarks. We use benchmarks with published protocols and ANN results. **BSuite memory length** [32] shows a cue, then L uninformative steps, and rewards the agent for recalling the cue; we run $L \in \{8, 16, 32, 64\}$, and $L \in \{51, 71, 101\}$ with the lengths, budget (1M steps) and training windows of R2I [38]. **POPGym RepeatPrevious** [26] rewards the agent at every step for naming the suit of the card shown k steps earlier ($k = 4$ Easy, $k = 32$ Medium), one of the three POPGym tasks with the highest memory demand that R2I evaluates; we use the environment settings of R2I's code for 2M steps. **Memory Maze 9×9** [33] is first-person 3D navigation with episodes of 1000 steps; we use its offline probing benchmark with the official dataset and probes: after 1M world model updates with truncated backpropagation through time, probes decode the positions of the target objects (MSE) and the maze layout (accuracy) from the frozen state. **DMC** [43] from proprioception (walker walk, cheetah run, cartpole swingup; 500k steps, train ratio 512, as in DreamerV3) and walker walk from pixels (1M frames) serve as control tasks without a memory demand. On BSuite the reward depends on both the cue and the action, which the scalar reward head of a Dreamer world model does not learn within our budgets; all models therefore use a reward head with one output per action (Supp. B). Supp. A gives the settings and the evaluation protocol of each benchmark.

World model probes. To measure memory without exploration and credit assignment, we also train world models alone on episodes of a random policy and measure the AUC of their reward prediction where the reward depends on the past, from 0.5 without memory to 1 with perfect recall: cue recall, BSuite memory length with the terminal reward set to the cue L steps back (± 1), and RepeatPrevious, where the reward indicates whether the action names the card k steps back. As in probing benchmarks [33], we report the final AUC after a fixed number of updates (Supp. C).

5.2. Memory in reinforcement learning

Table 1, Fig. 1 and Fig. 3 show how far each model remembers. The spiking baselines remember as far as their neurons keep their state. Spiking-WM, cleared at every step and at every spike, recalls no cue, not even 8 steps back, and recurrent LIF neurons, cleared at every step, solve 8 steps. GRSN and DH-SNN carry leaky states across steps and solve 16 steps. TC-LIF adds its input to two compartments that do not decay, and its spikes subtract a fixed amount from them instead of clearing them; it solves 64 steps and, with R2I's lengths, 51 and 101 steps (71: in progress), as the design rule of Sec. 3.3 predicts. DreamerV3 solves up to 32 steps in the short budgets and 51 steps with R2I's lengths (after 902k of 1M steps), and fails at 64, 71 and 101. **[TODO: Recurrent LIF, GRSN and DH-SNN at 51 steps (runs in**

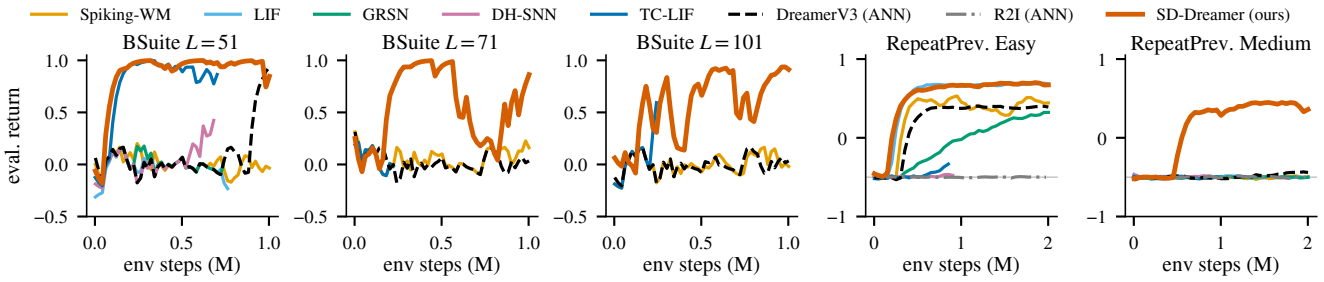


Figure 3. **Learning curves** on BSuite memory length with the lengths and budget of R2I (left three) and POPGym RepeatPrevious (right two; chance $= -0.5$). Exponential moving average of the evaluation return (R2I: return of the training episodes, see Tab. 1). Runs in progress.

Table 2. **World model probes** (data of a random policy, no policy). Final reward AUC (0.5 = no memory), mean of the last three evaluations after 12k (cue recall) or 10k (RepeatPrevious) updates; updates until 0.9 in Supp. C. *Held out (used for no design choice): 20k updates, training window 128. Blue: in progress.

	cue recall				RepeatPrevious		
	L=8	16	32	64	k=4	32	64*
Spiking-WM [41]	0.49	0.54	0.66	0.52	0.97	0.50	0.50
Recurrent LIF	1.00	0.52	0.50	0.50	0.99	0.50	0.50
GRSN [35]	1.00	1.00	0.51	0.51	0.89	0.51	0.50
DH-SNN [52]	1.00	1.00	1.00	0.52	0.63	0.50	0.50
TC-LIF [50]	1.00	1.00	1.00	1.00	0.86	0.50	0.57
DreamerV3 (ANN) [15]	1.00	1.00	1.00	0.50	0.95	0.50	0.50
SD-Dreamer (ours)	1.00	1.00	1.00	1.00	0.98	0.92	0.94

progress).]

SD-Dreamer reaches a return of 0.9 at every length, as fast as the fastest baseline or faster: 64 steps after 82k environment steps (TC-LIF 112k), and with R2I's lengths 51, 71 and 101 steps after 62k, 182k and 142k (TC-LIF 102k, >222k and 242k; returns in Supp. A). R2I reports a median return of about 0.9 at 51 and 71 memory steps and about 0.35 at 101 after 1M steps (10 seeds; Fig. 7 of [38]), with DreamerV3's default reward head and lengths that count one step less.

On RepeatPrevious Easy, SD-Dreamer reaches 0.72, close to the best baseline, recurrent LIF (0.74). On Medium, which asks for the card of exactly 32 steps ago, every baseline, DreamerV3 included, stays at chance after 2M steps, while SD-Dreamer reaches 0.48. R2I, run with its official code, stays at chance on both tasks within 2M steps. In its published curves, R2I leaves chance on Medium only after about 3M steps, and DreamerV3 learns Easy but not Medium (Fig. 9 of [38]), as in our runs. **[TODO: DH-SNN and TC-LIF on POPGym (runs in progress).]**

5.3. What the world model remembers

The probes separate memory from policy learning (Tab. 2). On cue recall, SD-Dreamer and TC-LIF recall 64 steps (AUC 1.00); DH-SNN and DreamerV3 stop at 32, GRSN at 16, recurrent LIF at 8, and Spiking-WM recalls no length. The order is that of RL, so the limits of the baselines are limits of their world models.

RepeatPrevious asks for a different memory: the input of an exact lag among a stream of similar inputs. At $k = 4$ most models learn it (AUC 0.86 to 0.99; SD-Dreamer 0.98). At $k = 32$, SD-Dreamer is the only model that learns it (AUC 0.92); all others, TC-LIF and DreamerV3 included, stay at 0.50 to 0.51. Resonance is the reason: without resonant channels the same model reaches only 0.57 (Tab. 5). TC-LIF recalls the cue 64 steps back as well as SD-Dreamer, but its compartments oscillate without decay at one period of about 12 steps that all neurons share (Supp. A), whereas each resonant channel of SD-Dreamer has its own phase and retention time. We added resonant channels after an earlier model with integrator channels only failed this task (Supp. F) and used RepeatPrevious Hard ($k = 64$) for no design choice. On Hard, SD-Dreamer reaches an AUC of 0.94, and all other models stay at 0.50 to 0.57.

Figure 4 shows where this memory lives, in the world models of the agents on BSuite memory length 101. When two episodes differ only in the cue, the spikes of Spiking-WM become identical 10 steps after the cue, since no state outlasts the step reset. In SD-Dreamer, the dendritic channels keep the difference for all 101 steps, and so do the spikes of the soma (3% of the firing rate at the query). Integrator channels with long retention times hold the difference with a constant sign, and each resonant channel rotates it with its own period, so that its phase encodes the time since the cue.

5.4. Memory Maze

Memory Maze tests whether a world model builds a map of a 3D maze from first-person views over 1000 steps. Table 3 reports the probes on the state that the policy reads: (z_t, h_t) for all models, and in addition the dendritic spikes for SD-

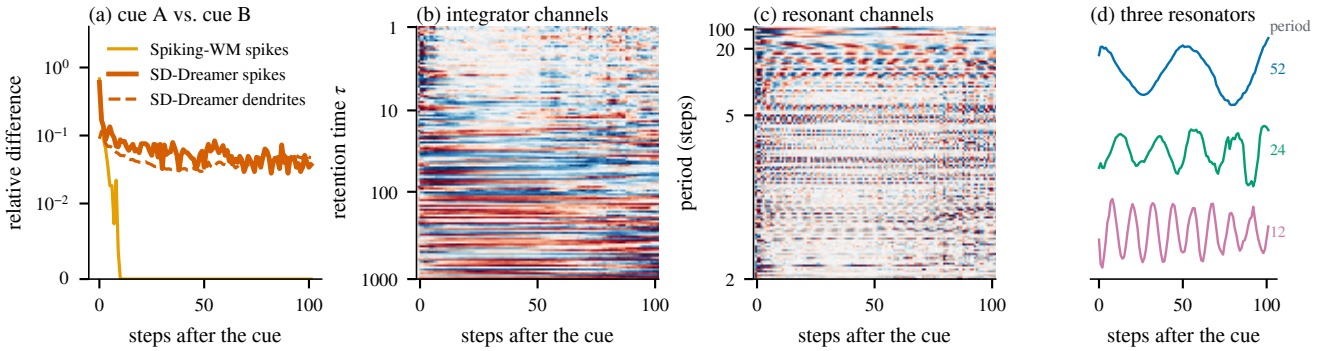


Figure 4. **Where the cue is held** (BSuite memory length 101, world models of the agents of Tab. 1). Two evaluation episodes that differ only in the cue enter each world model, with the stochastic state at its mode, so that the two runs differ only through the cue. (a) Difference between the two episodes, relative to the mean, of the spikes that carry the recurrent state to the next step (Spiking-WM: its recurrent core; SD-Dreamer: the second PDN layer) and of the dendritic channels of SD-Dreamer. (b, c) Difference of each channel of the second PDN layer, scaled to the largest value of the channel: integrator channels sorted by retention time, resonant channels by period $2\pi/\theta$. (d) Three resonant channels with retention times above 200 steps; numbers: period in steps.

Table 3. **Memory Maze and DMC.** Memory Maze 9×9 offline probing [33]: probes on the frozen state after 1M updates, second half of the evaluation episodes (constant predictor: 23.9 MSE, 80.8% walls). DMC from proprioception: return after 500k steps; walker walk from pixels: return after 1M frames. n/a: not run. Bottom: reference results from [33] (larger RSSM).

	Memory Maze 9×9		DMC, proprio. ↑			pixels ↑
	obj. MSE ↓	walls % ↑	walk	run	swing	walk
Spiking-WM [41]	11.9	83.7	923	343	497	917
Recurrent LIF	11.7	83.7	807	532	811	478
GRSN [35]	11.5	83.3	863	392	652	389
DH-SNN [52]	n/a	n/a	n/a	n/a	n/a	417
TC-LIF [50]	n/a	n/a	n/a	n/a	n/a	457
DreamerV3 (ANN) [15]	9.8	85.3	953	699	871	968
SD-Dreamer (ours)	9.4	82.8	923	487	676	842
RSSM [33]	8.3	89.0	n/a	n/a	n/a	n/a
RSSM, TBTT [33]	6.2	94.0	n/a	n/a	n/a	n/a

Table 4. **Interventions on Spiking-WM** (world model probe, cue recall over L steps). Updates until the reward AUC exceeds 0.9, two seeds; – = not within 12k updates; blank = not run. A: state kept across environment steps; B: no dendritic reset on a somatic spike; G: dendritic input scaled by $\sqrt{\tau_d/2}$. GRU: reference core with continuous state.

	A	B	G	τ_d	$L=8$	$L=16$	$L=32$
Spiking-WM				2	– / –	– / –	12k / –
				16	– / –	– / –	
				128	– / –	– / –	
B only		✓		128	– / –		
G only			✓	128		– / –	
A	✓			2	– / 9k	– / –	
	✓			128	3k / 4k	8k / 8.5k	
A+B	✓	✓		128	3k / 3k	4.5k / 4.5k	– / –
A+B+G	✓	✓	✓	128	**0.5k / 0.5k**	**0.5k / 0.5k**	**2k / 3k**
GRU core	continuous state				0.5k / 0.5k	1k / 1k	5k / 3.5k

Dreamer. SD-Dreamer locates the target objects as well as DreamerV3 (MSE 9.8) and better than every spiking baseline (11.5 to 11.8). On the wall layout, all models of this size stay within 5 points of the constant predictor (80.8%), SD-Dreamer at 82.8% and DreamerV3 at 85.3%; the larger RSSM of [33] reaches 89.0%. Figure 5 shows what this memory does. While an object is in view, every model locates it (squared error 1.4 to 3.5). Once the object leaves the view, the error of the spiking baselines rises within 3 to 5 steps to 7.3 to 11.7, whereas SD-Dreamer stays at 3.8 and has the lowest error of all models, DreamerV3 included, from 6 to 100 steps after the object left the view. **[TODO: Memory Maze: final probes after 1M updates (in progress); the values are the latest probes during training.]**

5.5. Causes of reset amnesia

We test the three mechanisms of Sec. 3.3 with interventions on Spiking-WM in the cue recall probe (Tab. 4; details in Supp. C). **A** keeps all compartments across environment steps and clears them only at episode starts, **B** keeps the dendrites when the soma fires, and **G** multiplies the dendritic input by $\sqrt{\tau_d/2}$, which equalizes the stationary spread across dendritic time constants τ_d.

With resets, time constants do not matter. Spiking-WM fails at 8 and 16 steps for $\tau_d \in \{2, 16, 128\}$, and the cue can be decoded from its spikes for only 2 to 3 steps after it is shown. At 32 steps, one of two seeds crosses an AUC of 0.9 at the last of 12k updates by holding the cue in self-sustained recurrent firing, a memory that exists only as spikes. **Persis-**

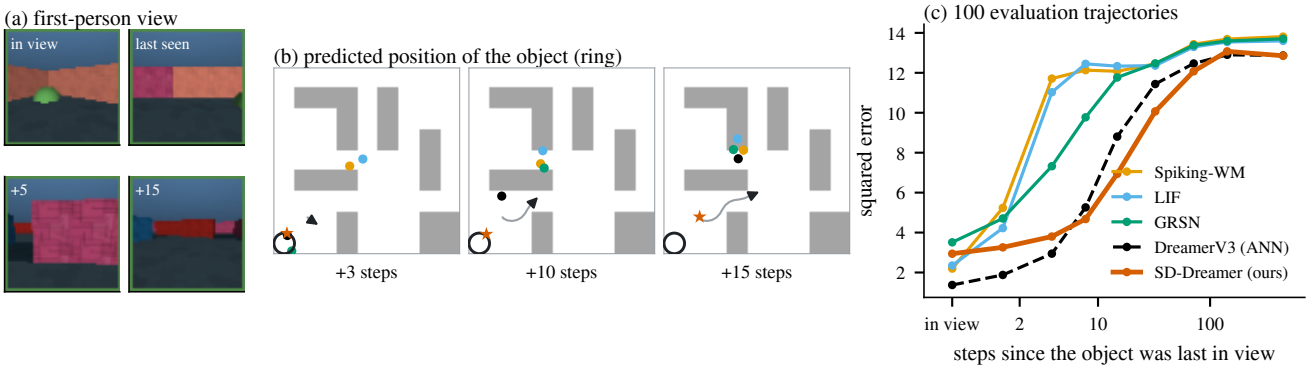


Figure 5. **Memory of objects out of view** (Memory Maze 9×9, the probes of Tab. 3). (a) First-person views around the first time, in evaluation trajectory 0, that an object in view for 5 steps leaves the view for at least 40 steps. (b) The position of this object (ring) that the probe of each model predicts 3, 10 and 15 steps later (SD-Dreamer: star); arrow: the agent; grey: its path since it last saw the object. (c) Squared error of the predicted object position against the steps since the object was last in view (within the 90° field of view, no wall in between), first 100 evaluation trajectories. SD-Dreamer: checkpoint after 330k of 1M updates.

tence is necessary. Without A, neither B nor G helps. With A and $\tau_d = 128$, the world model recalls 8 and 16 steps; removing the dendritic reset too (A+B) nearly halves the updates at 16 steps (4.5k instead of 8k to 8.5k). **The write gain limits long time constants.** With both resets removed, a cue moves the basal potential of a $\tau_d = 128$ dendrite by 0.004 to 0.009, about 1% of the threshold, against 0.19 at $\tau_d = 2$. A+B therefore fails at 32 steps, although a linear probe still decodes the cue from the membrane potentials at the recall step (accuracy 0.88): the trace is there, but too faint to drive spikes. Restoring the gain (A+B+G) recalls 8 and 16 steps within 500 updates and 32 steps within 2000 to 3000, faster than the GRU core, and still learns 32 steps in 2000 updates with the firing rate capped at 0.06. **Longer memory needs more than removing resets.** A+B+G fails at 64 steps even with $\tau_d = 512$, and in reinforcement learning it stops at 32 steps (Spiking-WM A+B+G in Supp. A). Its time constants are counted in simulation steps, and its soma still resets; a PDN counts retention times in environment steps and has a stateless soma.

Table 5. **Ablations** on the world model probes (as in Tab. 2). Each row changes one design choice of SD-Dreamer.

	cue recall		RepeatPrev.	firing
variant	L=32	64	k=32	rate
SD-Dreamer	1.00	1.00	0.92	0.25
no PDM (cleared every step)	0.51	0.50	0.50	0.25
single $\tau = 16$	1.00	0.49	0.92	0.24
single $\tau = 128$	1.00	1.00	0.93	0.26
τ fixed at initialization	1.00	1.00	0.93	0.24
no NWG (Euler gain $1 - \lambda$)	1.00	0.52	0.76	0.21
one PDN layer	1.00	0.50	0.91	0.21
one PDN layer, $T = 1$	0.99	0.99	0.50	0.15
no resonant channels	1.00	1.00	0.57	0.27
all channels resonant	1.00	1.00	0.95	0.23
firing rate capped at 0.06	1.00	0.51	0.92	0.06
soma with membrane and reset	1.00	0.52	0.77	0.18
Heaviside soma	1.00	1.00	0.88	0.16
$M = 4$ channels	1.00	1.00	0.93	0.25
no spike feedback	1.00	1.00	0.94	0.23

5.6. Ablations

Table 5 changes one design choice at a time (updates until 0.9 in Supp. C). **PDM**: clearing the dendrites at every environment step erases the memory (AUC 0.50 to 0.51 on all three tasks), the direct test of reset amnesia inside our own model. **Retention time**: a single retention time of 16 steps fails at 64 steps and one of 128 succeeds; the spectrum from 1 to 1000 steps succeeds without being set for the task, learned or kept at its initialization. **NWG**: with the Euler gain $1 - \lambda$ the channels with long retention times are too faint, and the model fails at 64 steps (0.52) and reaches only 0.76 at $k = 32$. **Stateless soma**: a soma that keeps a membrane potential and is reset by its spike fails at 64 steps (0.52) and reaches 0.77 at $k = 32$. **Resonance**: without resonant channels the model fails at $k = 32$ (0.57); with resonant channels only it solves all three tasks, as does SD-Dreamer with one channel of each kind. **Depth and sparsity**: one PDN layer, or a firing rate capped at 0.06 (from 0.25), fails at 64 steps. The spike count, the feedback and the number of channels leave the memory nearly unchanged (AUC 0.88 to 0.94 at $k = 32$); the spike count and the feedback serve the prediction of motion: on the same pixel walker episodes, the world model with both predicts the next frame as well as recurrent LIF neurons (image loss 44.3 vs. 44.1), and with one spike per step it does not (55.3; Supp. F).

5.7. Control tasks

On DMC from proprioception (Tab. 3), SD-Dreamer matches the best spiking model on walker walk (923, as Spiking-WM) and reaches 487 on cheetah run and 676 on cartpole swingup, where recurrent LIF reaches 532 and 811. An actor that reads (z_t, h_t) only reaches the same return with the same world model on walker walk and cheetah run (904 and 473) and a higher one on cartpole swingup (813; Supp. E). **[TODO: Walker walk from pixels after 1M frames (runs in progress).]** From pixels, every world model predicts the walker from the actions alone for about 10 to 20 steps (Fig. 6); the frames that SD-Dreamer predicts stay closer to the observed ones than those of Spiking-WM at all 45 steps, and than those of DreamerV3 from step 15 on.

6. Conclusion

Existing spiking world models forget because their neurons are cleared at every environment step and at every spike, which leaves their time constants nothing to act on. SD-Dreamer is built on persistent dendritic neurons: PDM keeps their integrator and resonant dendrites through environment steps and somatic spikes without reset, NWG writes into channels of every retention time on one scale, and a stateless soma communicates the memory by spike counts. It recalls cues 101 steps back on BSuite within the budget of R2I and is the only model in our comparison that recalls exact lags of 32 and 64 steps.

Limitations. The dendritic state is graded, so deployment needs neuromorphic cores whose neurons hold several state variables and that compute layer normalization.

References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016. 4

[2] Malyaban Bal and Abhronil Sengupta. P-SpikeSSM: Harnessing probabilistic spiking state space models for long-range dependency tasks. In *International Conference on Learning Representations (ICLR)*, 2025. 3

[3] Guillaume Bellec, Darjan Salaj, Anand Subramoney, Robert Legenstein, and Wolfgang Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 1, 2, 3

[4] David Beniaguev, Idan Segev, and Michael London. Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17):2727–2739, 2021. 3

[5] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1):82–99, 2018. 1

[6] Fei Deng, Junyeong Park, and Sungjin Ahn. Facing off world model backbones: RNNs, transformers, and S4. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2, 3

[7] Maxime Fabre, Lyubov Dudchenko, Younes Bouhadjar, and Emre Neftci. SiLIF: Structured state space model dynamics and parametrization for spiking neural networks. *arXiv preprint arXiv:2506.06374*, 2025. 3

[8] Wei Fang, Zhaofei Yu, Yanqi Chen, Tiejun Huang, Timothée Masquelier, and Yonghong Tian. Deep residual learning in spiking neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 1

[9] Wei Fang, Zhaofei Yu, Yanqi Chen, Timothée Masquelier, Tiejun Huang, and Yonghong Tian. Incorporating learnable membrane time constant to enhance learning of spiking neural networks. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2641–2651, 2021. 3

[10] Wei Fang, Zhaofei Yu, Zhaokun Zhou, Ding Chen, Yanqi Chen, Zhengyu Ma, Timothée Masquelier, and Yonghong Tian. Parallel spiking neurons with high efficiency and ability to learn long-term dependencies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[11] David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 2

[12] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning (ICML)*, 2019. 2, 3

[13] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020. 2

[14] Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering Atari with discrete world models. In *International Conference on Learning Representations (ICLR)*, 2021.

[15] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. 2, 3, 4, 6, 7, 8

[16] Saya Higuchi, Sebastian Kairat, Sander M. Bohte, and Sebastian Otte. Balanced resonate-and-fire neurons. In *International Conference on Machine Learning (ICML)*, 2024. 3

[17] Justus Huebotter, Pablo Lanillos, Marcel van Gerven, and Serge Thill. Spiking neural networks for continuous control via end-to-end model-based learning. *arXiv preprint arXiv:2509.05356*, 2025. 3

[18] Bruce Hutcheon and Yosef Yarom. Resonance, oscillation and the intrinsic frequency preferences of neurons. *Trends in Neurosciences*, 23(5):216–222, 2000. 3

[19] Eugene M. Izhikevich. Resonate-and-fire neurons. *Neural Networks*, 14(6–7):883–894, 2001. 2, 3, 4

[20] Matthew Larkum. A cellular mechanism for cortical associations: An organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3):141–151, 2013. 3

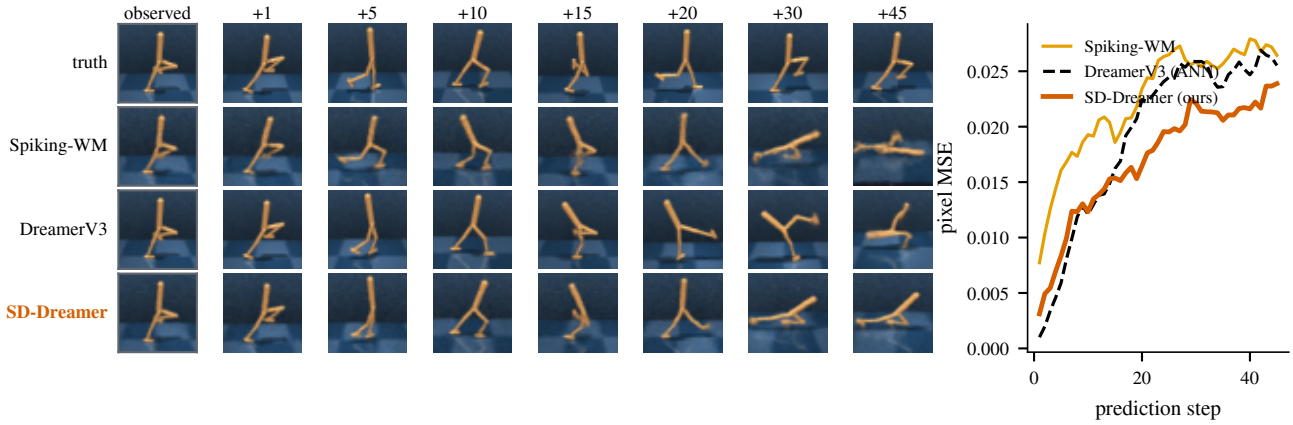


Figure 6. **Open-loop prediction from pixels** (DMC walker walk). Each world model observes 5 frames of the same evaluation episode of SD-Dreamer and then predicts 45 frames from the actions alone. Right: pixel MSE of the predicted frames, mean over 8 segments of the episode. SD-Dreamer: checkpoint after 448k of 1M frames.

[21] Guisong Liu, Wenjie Deng, Xiurui Xie, Li Huang, and Huajin Tang. Human-level control through directly trained deep spiking Q-networks. *IEEE Transactions on Cybernetics*, 53 (11):7187–7198, 2023. 1, 2, 3, 4

[22] Michael London and Michael Häusser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. 3

[23] Chris Lu, Yannick Schroecker, Albert Gu, Emilio Parisotto, Jakob Foerster, Satinder Singh, and Feryal Behbahani. Structured state space models for in-context reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[24] Wolfgang Maass. Networks of spiking neurons: The third generation of neural network models. *Neural Networks*, 10 (9):1659–1671, 1997. 1, 3

[25] Vincent Micheli, Eloi Alonso, and François Fleuret. Transformers are sample-efficient world models. In *International Conference on Learning Representations (ICLR)*, 2023. 3

[26] Steven Morad, Ryan Kortvelesy, Matteo Bettini, Stephan Liwicki, and Amanda Prorok. POPGym: Benchmarking partially observable reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2023. 2, 3, 6

[27] Steven Morad, Ryan Kortvelesy, Stephan Liwicki, and Amanda Prorok. Reinforcement learning with fast and forgetful memory. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[28] Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019. 3

[29] Tianwei Ni, Michel Ma, Benjamin Eysenbach, and Pierre-Luc Bacon. When do transformers shine in RL? decoupling memory from credit assignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[30] Garrick Orchard, E. Paxon Frady, Daniel Ben Dayan Rubin, Sophia Sanborn, Sumit Bam Shrestha, Friedrich T. Sommer, and Mike Davies. Efficient neuromorphic signal processing with Loihi 2. In *IEEE Workshop on Signal Processing Systems (SiPS)*, pages 254–259, 2021. 1

[31] Antonio Orvieto, Samuel L. Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, and Soham De. Resurrecting recurrent neural networks for long sequences. In *International Conference on Machine Learning (ICML)*, 2023. 3, 5

[32] Ian Osband, Yotam Doron, Matteo Hessel, John Aslanides, Eren Sezener, Andre Saraiva, Katrina McKinney, Tor Lattimore, Csaba Szepesvári, Satinder Singh, Benjamin Van Roy, Richard Sutton, David Silver, and Hado van Hasselt. Behaviour suite for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2020. 1, 2, 3, 6

[33] Jurgis Pasukonis, Timothy Lillicrap, and Danijar Hafner. Evaluating long-term memory in 3D mazes. In *International Conference on Learning Representations (ICLR)*, 2023. 2, 3, 6, 8

[34] Panayiota Poirazi, Terrence Brannon, and Bartlett W. Mel. Pyramidal neuron as two-layer neural network. *Neuron*, 37 (6):989–999, 2003. 3

[35] Lang Qin, Ziming Wang, Runhao Jiang, Rui Yan, and Huajin Tang. GRSN: Gated recurrent spiking neurons for POMDPs and MARL. In *AAAI Conference on Artificial Intelligence*, pages 1483–1491, 2025. 1, 3, 5, 6, 7, 8

[36] Jan Robine, Marc Höftmann, Tobias Uelwer, and Stefan Harmeling. Transformer-based world models are happy with 100k interactions. In *International Conference on Learning Representations (ICLR)*, 2023. 3

[37] Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575:607–617, 2019. 1

[38] Mohammad Reza Samsami, Artem Zholus, Janarthanan Rajendran, and Sarath Chandar. Mastering memory tasks with world models. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 2, 3, 6, 7

[39] Shuaijie Shen, Chao Wang, Renzhuo Huang, Yan Zhong, Qinghai Guo, Zhichao Lu, Jianguo Zhang, and Luziwei Leng. SpikingSSMs: Learning long sequences with sparse and parallel spiking state space models. In *AAAI Conference on Artificial Intelligence*, pages 20380–20388, 2025. 3

[40] Matei-Ioan Stan and Oliver Rhodes. Learning long sequences in spiking neural networks. *Scientific Reports*, 14:21957, 2024. 3

[41] Yinqian Sun, Feifei Zhao, Mingyang Lyu, and Yi Zeng. Spiking world model with multicompartment neurons for model-based reinforcement learning. *Proceedings of the National Academy of Sciences*, 122(50):e2513319122, 2025. 2, 3, 4, 5, 6, 7, 8

[42] Guangzhi Tang, Neelesh Kumar, Raymond Yoo, and Konstantinos P. Michmizos. Deep reinforcement learning with population-coded spiking neural network for continuous control. In *Conference on Robot Learning (CoRL)*, 2020. 1, 2, 3, 4

[43] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018. 2, 6

[44] Aaron Voelker, Ivana Kajić, and Chris Eliasmith. Legendre memory units: Continuous-time representation in recurrent neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 3

[45] Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari S. Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023. 3

[46] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in Neuroscience*, 12:331, 2018. 3

[47] Man Yao, Jiakui Hu, Zhaokun Zhou, Li Yuan, Yonghong Tian, Bo Xu, and Guoqi Li. Spike-driven transformer. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1

[48] Bojian Yin, Federico Corradi, and Sander M. Bohté. Accurate and efficient time-domain classification with adaptive spiking recurrent neural networks. *Nature Machine Intelligence*, 3 (10):905–913, 2021. 1, 3

[49] Dehao Zhang, Malu Zhang, Shuai Wang, Jingya Wang, Wenjie Wei, Zeyu Ma, Guoqing Wang, Yang Yang, and Haizhou Li. Dendritic resonate-and-fire neuron for effective and efficient long sequence modeling. *arXiv preprint arXiv:2509.17186*, 2025. 3

[50] Shimin Zhang, Qu Yang, Chenxiang Ma, Jibin Wu, Haizhou Li, and Kay Chen Tan. TC-LIF: A two-compartment spiking neuron model for long-term sequential modelling. In *AAAI Conference on Artificial Intelligence*, pages 16838–16847, 2024. 1, 2, 3, 6, 7, 8

[51] Weipu Zhang, Gang Wang, Jian Sun, Yetian Yuan, and Gao Huang. STORM: Efficient stochastic transformer based world models for reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[52] Hanle Zheng, Zhong Zheng, Rui Hu, Bo Xiao, Yujie Wu, Fangwen Yu, Xue Liu, Guoqi Li, and Lei Deng. Temporal dendritic heterogeneity incorporated with spiking neural networks for learning multi-timescale dynamics. *Nature Communications*, 15:277, 2024. 1, 2, 3, 5, 6, 7, 8

[53] Zhaokun Zhou, Yuesheng Zhu, Chao He, Yaowei Wang, Shuicheng Yan, Yonghong Tian, and Li Yuan. Spikformer: When spiking neural network meets transformer. In *International Conference on Learning Representations (ICLR)*, 2023. 1